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A comparative study on oxidative stress response in the hepatopancreas and midgut of the white shrimp *Litopenaeus vannamei* under gradual changes to low or high pH environment

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ABSTRACT

White shrimp *Litopenaeus vannamei* were reared under conditions of gradual changes to a low pH (gradual-low pH, 6.65–8.20) or a high pH (gradual-high pH, 8.20–9.81) versus a normal pH environment (8.14–8.31) during a 28-day period. Survival of shrimp, and ROS production, antioxidant responses and oxidative damage in the hepatopancreas and midgut were investigated. Consequently, shrimp enhanced *MnSOD*, *GPx*, and *Hsp70* transcripts as early defense mechanism in the hepatopancreas and midgut to scavenge excessive ROS during short-term (≤ 7 d) gradual-low and high pH stress. Meanwhile, the hepatopancreas was more sensitive to ROS than midgut because of earlier ROS production increase, antioxidant response and oxidative damage. Then, suppressed antioxidant response in the hepatopancreas and midgut of shrimp suggested a loss of antioxidant regulatory capacity caused by aggravated oxidative damage after long-term (≥ 14 days) gradual-high pH stress, leading to continuous death. However, enhanced *GPx*, *GST*, and *Hsp70* transcripts in the hepatopancreas and midgut might be long-term (≥ 14 days) antioxidant adaptation mechanism of shrimp to gradual-low pH stress, which could prevent further ROS perturbation and weaken oxidative damage to achieve a new immune homeostasis, contributing to stable survival rate. Therefore, we have a few insights that it is necessary to protect hepatopancreas for controlling shrimp death under gradual-high pH stress.

Keywords: *Litopenaeus vannamei*, hepatopancreas, midgut, gradual-low pH, gradual-high pH, antioxidant

1. Introduction

In aquaculture ponds, pH levels tend to fluctuate from 6.6 to 10.2 due to the consumption and release of carbon dioxide [1]. Accumulated high concentrations of soluble organic substances in ponds may promote the formation of a red tide, leading to the water pH up to 9.0 [2]. Conversely, the sediments in earth ponds can induce water acidification [3]. Euryhaline shrimp are a taxonomic group of aquatic crustaceans that can tolerate a wide range of salinity. For these organisms, pH is an important factor affecting many physiological functions. It has been reported that low or high pH could induce oxidative stress, reduce immunity parameters, and retard feeding, molting, and growth of shrimp, leading to decreased resistance against *Vibrio*, and even severe death [4-7]. As the hepatopancreas and intestine in shrimp are key organs involved in digestion, detoxification and immunity [8-10], their defense responses for the endurance of shrimp to low or high pH environment might cause concern, and comparative study on oxidative stress response in these vital organs is necessary to develop an effective strategy for controlling shrimp death and subsequent economic loss under low or high pH environment.

The release of reactive oxygen species (ROS) is a crucial step in the innate immune to eliminate potential pathogens and parasites following phagocytosis in normal cell functions, but excessive ROS will also cause potential cytotoxic problem [11,12]. When the organism encounters environmental stress, ROS including superoxide anion (O_2^-), hydroxyl radical, and hydrogen peroxide (H_2O_2) are released by oxidative stress [13, 14]. O_2^- is the first product during this process, and they are continuously generated by an aerobic metabolism and can damage important biomolecules, such as DNA, proteins, and lipids [15]. Malondialdehyde (MDA) as the main component of lipid peroxides, has a strong biotoxicity and will damage the cell structure and function [16]. The ROS, MDA could attack the DNA in cell at the same time or individually, leading to DNA damage. Subsequently, multicellular animals would acquire the ability to eliminate damaged cells by triggering apoptosis [17].

To counteract ROS-induced damage for stress tolerance, cells have evolved antioxidant enzyme system as well-developed regulatory mechanism to prevent or

repair such damage [18,19]. Superoxide dismutase (SOD) catalyzes O_2^- to H_2O_2 and molecular oxygen (O_2) [20]. Glutathione peroxidase (GPx) catalyze the dismutation of H_2O_2 to water and O_2 . Concomitantly, glutathione-S-transferase (GST) facilitates the elimination of toxicants and products of oxidative damage [21,22]. In addition to the antioxidant enzymes, Heat shock proteins (HSPs) as stress proteins and extrinsic chaperones are also important components of various antioxidant and detoxification pathways [23]. Particularly, Hsp70 could help the cell to prevent or repair protein misfolding from ROS-induced damage [24,25]. Thus, both antioxidant enzymes and chaperone proteins play important roles in antioxidant responses, maintaining cellular immune homeostasis [26].

The white shrimp *Litopenaeus vannamei* is the most widely cultured euryhaline shrimp species in the world. Therefore, we evaluated survival, and investigated (1) ROS production, (2) antioxidant responses (*MnSOD*, *GPx*, *GST*, and *Hsp70* gene expressions), and (3) oxidative damage (MDA content and apoptosis) in the hepatopancreas and midgut of *L. vannamei* reared under conditions of gradual changes to a low pH environment (gradual-low pH) or a high pH environment (gradual-high pH) versus a normal pH environment during a 28-day period. This study aimed to compare oxidative stress response in the hepatopancreas and midgut of *L. vannamei* under gradual-low or high pH versus normal pH, which is necessary to develop an effective strategy for controlling shrimp death.

2. Materials and methods

2.1. Aquarium Conditions

In total, 1350 healthy juvenile *L. vannamei* of similar size (mean weight 1.20 ± 0.03 g) were obtained from the Ruizi Seafood Development Co. Ltd. (Qingdao, China), where the experiment was also conducted. Only shrimp in the inter-molt stage were used for the study. Shrimp were placed in nine 640-L cylindrical tanks with a net cover ($N = 150$ per tank). Each tank contained 500 L of aerated seawater (dissolved oxygen 5.4–6.5 mg/L). The initial seawater was unfiltered with pH 8.0–8.4, salinity 30–31‰, total ammonia 0.022–0.038 mg/L, nitrite 0.015–0.032 mg/L, and nitrate 0.120–0.205 mg/L at 28–32°C. Animals were acclimated for 2 weeks under a natural

photoperiod (12 h light: 12 h dark). The shrimp were fed three times daily with a commercial diet (41.52% crude protein, 7.42% lipid, and 12.03 % crude ash, supplied by Yantai Dale feed Co. Ltd, Shandong, China) at 07:00 a.m., 11:00 a.m., and 5:00 p.m., representing a daily feeding rate that was 10% of the weight of shrimp. The unfed feed and feces were removed with a siphon tube and 35% seawater was replaced once daily. Unfiltered seawater was prepared in three 1000-L cylindrical tanks to use for exchange daily.

2.2. Experimental design for pH challenge

Following acclimation, nine 640-L cylindrical tanks were randomly divided into 3 groups. Each group had three tanks for the following conditions: (1) gradual-low pH (GLH), (2) normal pH (NH), and (3) gradual-high pH (GHH). Shrimp were subject to experimental conditions for 28 days. The photoperiod, feeding, and waste disposal conditions were handled in exactly the same manner as during the acclimation period. Seawater of varying pH, pH 6.7, pH 8.2, and pH 9.7, was prepared in three 1000-L cylindrical tanks to use for water changes for the GLH, NH and GHH conditions, respectively. Seawater of pH 6.7 and pH 9.7 was obtained by adjusting the pH 8.2 seawater with 4 M HCl or 4 M NaOH (Tianjin Yongda Chemical Reagent Co. Ltd, Tianjin, China). The pH levels were monitored daily using a PHB-4 pH meter (Shanghai Yidian Scientific Instruments CO. Ltd, Shanghai, China). During the experimental period, the pH of the seawater of GLH decreased from 8.20 to 6.72 within 7 days, then stabilized at 6.65–6.78 afterward, the pH of the seawater of NH stabilized at 8.14–8.31, and the pH of the seawater of GHH increased from 8.20 to 9.67 within 7 days, then stabilized at 9.64–9.81 afterward (Fig. 1).

2.3. Measurements of survival and sampling procedures

The number of dead shrimp from each tank was recorded every 24 h during the experimental period. Shrimp were considered dead when they failed to move even when gently stimulated with a glass pipette. Dead shrimp were removed to prevent fouling. Survival was evaluated in terms of survival rate (SR). $SR (\%) = 100 (\text{initial shrimp number} - \text{cumulative death shrimp number}) / (\text{initial shrimp number})$. Twelve shrimp were randomly selected and removed from each tank before the experiment (0)

and on days 1, 3, 7, 14, 28 during the experimental period. The hepatopancreas and the midgut from these twelve shrimp were collected using sterilized scissors and forceps: three hepatopancreases and midguts were immediately placed into RNAlater (Applied Biosystems, Austin, TX, USA) and stored at -20°C until RNA isolation and gene expression analysis; nine hepatopancreases and midguts were immediately ground in liquid nitrogen and stored at -80°C until use in the ROS production, MDA content and DNA ladder assay.

2.4. ROS production and MDA content in the hepatopancreas and the midgut

Six hepatopancreas and midgut tissues for each tank were respectively mixed at a 1:5 ratio (w/v) with chilled Tris-hydrochloric acid buffer solution ($\text{pH } 7.6$, 10 mmol L^{-1}) and homogenized under ice-chilled conditions. The homogenate was centrifuged at 3500 rpm , 4°C for 10 min , and the supernatant was then collected for sample analysis. ROS production in the hepatopancreas and midgut were measured respectively according to 2',7'-dichlorofluorescein diacetate (DCFH-DA) method [27]. 200 mL supernatant were incubated with DCFH-DA for 30 min in darkness, and diluted with 200 mL Modified Alsevier's Solution to a final concentration of $1 \times 10^6 \text{ cells/mL}$, followed by the flow cytometry analysis. ROS production was expressed as mean fluorescence of DCF. MDA content was evaluated using commercial kits (Nanjing Jiancheng Bioengineering Institute) according to the manufacturer's instructions. MDA content was expressed as the relative concentration per milligram of soluble protein (nmol/mg protein). Both ROS production and MDA content were analyzed with three replicates of each sample.

2.5. Gene expression assays in the hepatopancreas and the midgut

RNA was extracted from hepatopancreas and midgut using an RNA fast extraction kit according to the manufacturer's instructions (Fastagen, Shanghai, China). RNA solutions from three hepatopancreases from each tank were mixed into $6 \mu\text{L}$, $2 \mu\text{L}$ RNA solution per hepatopancreas. RNA solutions from three midguts from each tank were mixed into $6 \mu\text{L}$, $2 \mu\text{L}$ RNA solution per midgut; every $6 \mu\text{L}$ of RNA solution was used for cDNA synthesis using a TransScript® One-Step gDNA Removal and cDNA Synthesis Kit according to the manufacturer's instructions (TransGen Biotech Co., Ltd,

Beijing, China). The relative expression of genes including *MnSOD*, *Gpx*, *GST* and *Hsp70* were evaluated by relative quantitative real-time PCR (qPCR) using TransStar Top Green qPCR Supermix according to the manufacturer's instructions (TransGen Biotech Co., Ltd). Each of the above genes was analyzed with three replicates of each sample. β -Actin was selected as a reference gene and specific primers were used for each gene (Table 1). qPCR was performed in the following two steps: denaturation at 94°C for 30 s, and then 40 cycles of 94°C for 5 s and 60°C for 30 s. The melting curve analysis was performed at the end of the qPCR to confirm the specificity of the PCR products, and relative expressions were determined using the $2^{-\Delta\Delta C_t}$ method [28].

2.6. DNA ladder assays in hepatopancreas and midgut

The DNA ladder assay was performed to evaluate apoptosis in cell [29]. DNA was extracted from the hepatopancreas and midgut using a TIANamp Marine Animals DNA kit according to the manufacturer's instructions (Tiangen, Beijing, China). The DNA solutions (equal volume as used for the conventional assay) were loaded in separate lanes, and electrophoresis was performed on 1% agarose gels. The results were analyzed using a gel imaging and analysis system (Beijing Youhua Zhaoqin Medical Equipment Co. Ltd., Beijing, China).

2.7. Statistical analysis

The data were all presented as the mean $\pm$ standard error (SE). Statistical analysis was performed with SPSS (version 17.0) (IBM, Armonk, NY, USA), and t-test was used to analyze differences between two experimental groups. The significance level was $P < 0.05$. All images were generated with Origin 8.6 software (OriginLab, Northampton, MA, America).

3. Results

3.1. Survival of shrimp reared at different pH levels

Compared with NH shrimp, the SR of both GLH and GHH shrimp were significantly ($P < 0.05$) lower on days 1, 3, 7, 14, 28; the SR of GLH shrimp reached its minimum of 93.33% on days 7, then stabilized during the period 7-28 days; the SR of GHH shrimp decrease with time until 60.1% on days 28 (Fig. 2).

3.2. ROS production of shrimp under gradual-low and high pH stress

Compared with NH shrimp, ROS production in the hepatopancreas of GLH and GHH shrimp significantly ($P < 0.05$) increased on days 1, 3, 7, 14, 28 (Fig. 3A). ROS production in the midgut of GLH and GHH shrimp significantly ($P < 0.05$) increased on days 3, 7, 14, 28 (Fig. 3B). However, ROS production in the hepatopancreas and midgut of GLH shrimp increased first and then declined, while GHH shrimp increased continuously (Fig. 3A, B).

3.3. *MnSOD* gene expression of shrimp under gradual-low and high pH stress

Compared with NH shrimp, *MnSOD* transcript in the hepatopancreas of GLH shrimp significantly ($P < 0.05$) increased on days 1, 3, then significantly ($P < 0.05$) decreased on days 7, 14, 28; *MnSOD* transcript in the hepatopancreas of GHH shrimp significantly ($P < 0.05$) increased on days 1, 3, then significantly ($P < 0.05$) decreased on days 14, 28 (Fig. 4A). Compared with NH shrimp, *MnSOD* transcript in the midgut of GLH shrimp significantly ($P < 0.05$) decreased on days 3, then significantly ($P < 0.05$) increased on days 7, 14, then significantly ($P < 0.05$) decreased on days 28; *MnSOD* transcript in the midgut of GHH shrimp significantly ($P < 0.05$) decreased on day 1, then significantly ($P < 0.05$) increased on days 3, 7, then significantly ($P < 0.05$) decreased on days 14, 28 (Fig. 4B).

3.4. *GPx* gene expression of shrimp under gradual-low and high pH stress

Compared with NH shrimp, *GPx* transcript in the hepatopancreas of GLH shrimp significantly ($P < 0.05$) increased on days 1, 3, 7, 14, 28; *GPx* transcript in the hepatopancreas of GHH shrimp significantly ($P < 0.05$) increased on days 3, 7, then significantly ($P < 0.05$) decreased on days 14, 28 (Fig. 5A). Compared with NH shrimp, *GPx* transcript in the midgut of GLH shrimp significantly ($P < 0.05$) decreased on day 1, then significantly ($P < 0.05$) increased on days 3, 7, 14, 28; *GPx* transcript in the midgut of GHH shrimp significantly ($P < 0.05$) decreased on days 3, then significantly ($P < 0.05$) increased on days 7, 14, then significantly ($P < 0.05$) decreased on day 28 (Fig. 5B).

3.5. *GST* gene expression of shrimp under gradual-low and high pH stress

Compared with NH shrimp, *GST* transcript in the hepatopancreas of GLH shrimp significantly ($P < 0.05$) decreased on days 1, 3, 7, 14, then significantly ($P < 0.05$)

increased on day 28; *GST* transcript in the hepatopancreas of GHH shrimp significantly ($P < 0.05$) decreased on days 1, 3, 7, 14, 28 (Fig. 6A). Compared with NH shrimp, *GST* transcript in the midgut of GLH shrimp significantly ($P < 0.05$) decreased on days 3, 7, then significantly ($P < 0.05$) increased on days 14, 28; *GST* transcript in the midgut of GHH shrimp significantly ($P < 0.05$) decreased on days 3, 7, 14, 28 (Fig. 6B).

3.6. *Hsp70* gene expression of shrimp under gradual-low and high pH stress

Compared with NH shrimp, *Hsp70* transcript in the hepatopancreas of GLH shrimp significantly ($P < 0.05$) increased on days 1, 14, 28; *Hsp70* transcript in the hepatopancreas of GHH shrimp significantly ($P < 0.05$) increased on days 1, 14, then significantly ($P < 0.05$) decreased on days 28 (Fig. 7A). Compared with NH shrimp, *Hsp70* transcript in the midgut of GLH shrimp significantly ($P < 0.05$) decreased on day 1, then significantly ($P < 0.05$) increased on days 3, 7, 14, then returned to normal on days 28. *Hsp70* transcript in the midgut of GHH shrimp significantly ($P < 0.05$) decreased on day 1, then significantly ($P < 0.05$) increased on days 3, 7, then significantly ($P < 0.05$) decreased on days 28 (Fig. 7B).

3.7. *MDA* content of shrimp under gradual-low and high pH stress

Compared with NH shrimp, *MDA* content in the hepatopancreas of GLH and GHH shrimp significantly ($P < 0.05$) increased on days 1, 3, 7, 14, 28. However, *MDA* content in the hepatopancreas of GLH shrimp increased first and then declined, while GHH shrimp increased continuously (Fig. 8A). Compared with NH shrimp, *MDA* content in the midgut of GLH shrimp significantly ($P < 0.05$) increased on days 3, 7, 14, then returned to normal on days 28; *MDA* content in the midgut of GHH shrimp significantly ($P < 0.05$) increased on days 3, 7, 14, 28 (Fig. 8B).

3.8. *DNA ladder* of shrimp under gradual-low and high pH stress

One of the standard characteristics of apoptosis is that the genome breaks into multiple 180–200 bp globule fragments of oligosomal nucleus [30]. Compared with NH shrimp, apoptosis was induced in the hepatopancreas of GLH shrimp on day 1, reached its maximum degree on days 14, and weakened afterward (Fig. 9A1, B1). Apoptosis was induced in the midgut of GLH shrimp on days 3, reached its maximum

degree on days 7, weakened on days 14, and disappeared on days 28 (Fig. 9A2, B2). Apoptosis was induced in the hepatopancreas of GHH shrimp on day 1, and in the midgut of GHH shrimp on days 3, then aggravated in a time-dependent manner afterward (Fig. 9B1, B2, C1, C2).

4. Discussion

ROS are generated during the progress of phagocytosis, which is activated by a respiratory burst [12]. Low and high pH stress have been reported to affect oxygen affinity and oxygen consumption of *Palaemonetes kadiakensis* [31]. Oxidative stress occurred in *L. vannamei* exposed to low and high pH stress, as indicated by increased ROS production in haemolymph [6,32]. Similarly, ROS production in the hepatopancreas and midgut of *L. vannamei* were also upregulated during 28-day gradual-low and high pH press. Hepatopancreas was regarded as the metabolic center for ROS production in crustacean [33]. Thus, it became relative easy to understand that earlier ROS production increase appeared in the hepatopancreas, rather than the midgut of shrimp in the present study. *L. vannamei* experienced significantly higher mortality after 24 h low or high pH stress [34]. Meanwhile, we also found continuous shrimp death with ascending ROS production in the hepatopancreas and midgut during 28-day gradual-high pH stress. However, the interesting thing was that shrimp obtained stable survival rate with descending ROS production in the hepatopancreas and midgut after 7-day gradual-low pH press. Some scholars even found no significant death and growth impairment of *L. vannamei* after 21-day low pH stress [35]. Therefore, we speculated that it might be related to the long-term antioxidant adaptation mechanism of shrimp to gradual-low pH.

In the course of normal metabolism in organisms, the production and elimination of ROS maintains a dynamic balance [36]. When aquatic organisms encounter environmental stress, a series of self-regulatory non-specific responses are initiated to reach a new dynamic balance [37]. One such response is upregulation of antioxidant enzymes and chaperone proteins, which constitute defense mechanisms against ROS to maintain immune homeostasis [24,38]. GSTs catalyse the conjugation of glutathione with compounds having electrophilic centers, generating less toxic and

more water-soluble products [39]. They have been used as biomarkers to estimate response to environmental chemicals [40,41], some examples of studies describe GST induction as a consequence of contaminant exposure [42,43]. However, *GST* transcript increased at 6 h and 12 h, and then decreased at 24 h in the hepatopancreas of *L. vannamei* under low and high pH [34]. *GST* transcript was also downregulated at 24 h in the hepatopancreas of *L. vannamei* under gradual-low and high pH. *GST* transcript was even downregulated at 6 h in the hepatopancreas of *L. vannamei* under high temperature [44]. Thus, high sensitivity of biomarker might promote *GST* transcript to have rapid variation under stress, which might be quickly suppressed with excessive ROS after detoxification. The stress response might also impact antioxidants (such as MnSOD, GPx, and Hsp70) to constitute the defence mechanism against ROS in aquatic animal [45,46]. *MnSOD* and *GPx* transcripts in the hepatopancreas of *L. vannamei* increased after 12 h low and high pH press [32]; *Hsp70* transcript in the hepatopancreas of *L. vannamei* increased after 4 h low and high pH stress [47]. Similarly, *MnSOD*, *GPx*, and *Hsp70* transcripts in the hepatopancreas and midgut of *L. vannamei* were also upregulated during short-term (≤ 7 d) gradual-low and high pH. Thus, we considered that the enhanced antioxidant response in the hepatopancreas and midgut of shrimp might be an early defense mechanism to scavenge excessive ROS under pH stress, protecting cellular component from oxidative damage. Meanwhile, it's easy to find earlier antioxidant response in the hepatopancreas, rather than the midgut of shrimp in the present study. This conclusion can be also perfectly interpreted from previous report that the hepatopancreas was the most sensitive tissue to ROS due to its high metabolic rate and its role in detoxification processes [8,48].

Exposure to environmental stress and elevation of ROS can also act synergistically to cause extensive lipid peroxidation and DNA damage leading to apoptosis as a cellular immune response [49,50]. MDA is the end product of lipid peroxidation, which is recognized as an important indicator of oxidative damage to the cellular membrane [51]. The DNA ladder assay described here is a simple, sensitive, cost-effective and rapid method for estimating apoptosis in single cells [52]. In the present study, lipid

peroxidation and apoptosis were induced in the hepatopancreas and midgut of shrimp during short-term (≤ 7 d) gradual-low and high pH stress. Similar to this result, MDA level increased in the hepatopancreas of *L. vannamei* after 3-day low and high pH stress [6]; apoptosis appeared in the hepatopancreas of *F. chinensis* after 12 h low and high pH stress [53]. Furthermore, earlier oxidative damage including lipid peroxidation and apoptosis in the hepatopancreas, rather than the midgut corroborated that the hepatopancreas was more sensitive to ROS than midgut under gradual-low and high pH stress.

Severe or persistent environmental stress may cause mutations in cell membrane structure and protein composition of the liver in aquatic animals. One of the known effects the animals responded to these stresses can be shown to stop transcription of *Hsps* [54]. Previous study verified that *Hsp70* transcript decreased in the liver of *Salmo trutta fario* and *Barbatula barbatula* exposed to water and sediment of the complexly polluted stream [55]. *Hsp70* transcript in the hepatopancreas of *Megalobrama amblycephala* decreased after 5-day high pH stress [56]. Similarly, *Hsp70* transcript was downregulated in the hepatopancreas and midgut of *L. vannamei* after 28-day gradual-high pH stress. *SOD*, *GPx* and *GST* transcripts in the hepatopancreas and midgut of *L. vannamei* were also downregulated after 14-day gradual-high pH stress, which is consistent with previous study that *SOD* activity in the haemolymph of *L. vannamei* decreased after 6-day high pH stress [57]. The decline of antioxidant responses was probably caused by the fact that the levels of ascending ROS production surpassed the clear threshold of their own [58]. Meanwhile, lipid peroxidation and apoptosis aggravated in the hepatopancreas and the midgut of shrimp after long-term (≥ 14 days) gradual-high pH stress. These results might indicate a loss of antioxidant regulatory capacity in the hepatopancreas and the midgut of shrimp after long-term (≥ 14 days) gradual-high pH stress, which was caused by aggravated oxidative damage. However, the different thing was that shrimp enhanced *GST*, *GPx*, and *Hsp70* transcripts in the hepatopancreas and the midgut after long-term (≥ 14 days) gradual-low pH stress. This result might be long-term antioxidant adaptation mechanism of shrimp to gradual-low pH stress, which

prevented further ROS perturbation and weakened lipid peroxidation and apoptosis in the hepatopancreas and the midgut of shrimp, even got rid of these oxidative damage in the midgut on days 28 to achieve a new immune homeostasis.

Conclusions

During short-term (≤ 7 d) gradual-low and high pH stress, enhanced *MnSOD*, *GPx*, and *Hsp70* transcripts in the hepatopancreas and midgut of *L. vannamei* might be an early defense mechanism used for excessive ROS scavenging to maintain immune homeostasis, and the hepatopancreas was more sensitive to ROS than midgut. Then, the hepatopancreas and midgut might lose antioxidant regulatory capacity after long-term (≥ 14 days) gradual-high pH stress, leading to continuous shrimp death. However, enhanced *GPx*, *GST*, and *Hsp70* transcripts in the hepatopancreas and the midgut might be long-term (≥ 14 days) antioxidant adaptation mechanism of shrimp to gradual-low pH stress, contributing to stable survival rate. Therefore, we have a few insights that it is necessary to protect hepatopancreas for controlling shrimp death under gradual-high pH stress.

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References

- [1] C.E. Boyd. Water quality in ponds for aquaculture, 1990.
- [2] A.L. Wang, W.N. Wang, The study of the relationship between red tide and environment factors in shrimp farming. *Acta Oceanol. Sin. (Chin.)* 17 (1995) 128–133.
- [3] H. Lemonnier, E. Bernard, E. Boglio, C. Goarant, J.C. Cochard, Influence of sediment characteristics on shrimp physiology: pH as principal effect. *Aquaculture* 240 (2004) 297–312.
- [4] W. Cheng, J.C. Chen, Effects of pH, temperature and salinity on immune parameters of the freshwater prawn *Macrobrachium rosenbergii*, *Fish Shellfish Immun.* 10 (2000) 387–391.
- [5] S.M. Chen, J.C. Chen, Effects of pH on survival, growth, molting and feeding of

- 391 giant freshwater prawn *Macrobrachium rosenbergii*. *Aquaculture* 218 (2003)
392 613–623.
- 393 [6] W.N. Wang, B.S. Li, J.J. Liu, L. Shi, M.J. Alam, S.J. Su, et al., The respiratory
394 burst activity and expression of catalase in white shrimp, *Litopenaeus vannamei*,
395 during long-term exposure to pH stress, *Ecotoxicology* 21 (2012) 1609–1616.
- 396 [7] Y.Y. Chen, J.C. Chen, K.C. Tseng, Y.C. Lin, C.L. Huang, Activation of immunity,
397 immune response, antioxidant ability, and resistance against *Vibrio alginolyticus* in
398 white shrimp *Litopenaeus vannamei* decrease under long-term culture at low pH. *Fish*
399 *Shellfish Immunol.* 46 (2015) 192–199.
- 400 [8] E.E. Ruppert, R.S. Fox, R.D. Barnes, *Invertebrate Zoology: A Functional Ecology*
401 *Approach*, Brooks Cole College Publishing, Orlando, FL, 2003.
- 402 [9] J. Robalino, J.S. Almeida, D. McKillen, J. Colglazier, H.F. Trent, Y.A. Chen, et al.,
403 Insights into the immune transcriptome of the shrimp *Litopenaeus vannamei*:
404 tissue-specific expression profiles and transcriptomic responses to immune
405 challenge, *Physiol. genomics* 29 (2007) 44–56.
- 406 [10] W. Soonthornchai, W. Rungrassamee, N. Karoonuthaisiri, P. Jarayabhand, S.
407 Klinbunga, K. Söderhäll, et al., Expression of immune-related genes in the digestive
408 organ of shrimp, *Penaeus monodon*, after an oral infection by *Vibrio harveyi*, *Dev.*
409 *Comp. Immunol.* 34 (2010) 19–28.
- 410 [11] T. Holmblad, K. Söderhäll, Cell adhesion molecules and antioxidative enzymes
411 in a crustacean, possible role in immunity, *Aquaculture* 172 (1999) 111–123.
- 412 [12] M. Muñoz, R. Cedeño, J. Rodriguez, K. Wpwwander, E. Mialhe, E. Bachère,
413 Measurement of reactive oxygen intermediate production in haemocytes of the
414 penaeid shrimp, *Penaeus vannamei*, *Aquaculture* 191 (2000) 89–107.
- 415 [13] B. Ranby, J.E. Rabek, Singlet Oxygen, In Wiley, Chichester England, 1978.
- 416 [14] Y. Morel, R. Barouki, Repression of gene expression by oxidative stress,
417 *Biochem J.* 342 (1999) 481–496.
- 418 [15] B. Halliwell, J.M.C. Gutteridge, Antioxidant defenses. In: Halliwell, B,
419 Gutteridge, JMC (Eds.), *Free Radicals in Biology and Medicine*, Oxford Univ. Press,
420 New York, 1999.

- [16] B.A. Freeman, J.D. Crapo, Biology of disease: free radicals and tissue injury, Lab. Investig. 47 (1982) 412–426.
- [17] W.J. Mai, J.L. Yan, L. Wang, Y. Zheng, Y. Xin, W.N. Wang, Acute acidic exposure induces p53-mediated oxidative stress and dna damage in tilapia (*oreochromis niloticus*) blood cells, Aquat. Toxicol. 100 (2010) 271–281.
- [18] J. Nordberg, E.S.T. Arnev, Reactive oxygen species, antioxidants, and mammalian thioredoxin system, Free Rad. Biol. Med. 31 (2001) 1287–1312.
- [19] A.G.A. Gomez, M.C.V. Barillas, U.A.B. Peregrino, L. Gupta, G.T. Gollas, L.J. Hernandez et al., The cytosolic manganese superoxide dismutase from the shrimp *Litopenaeus vannamei*, molecular cloning and expression, Dev. Comp. Immunol. 30 (2006) 893–900.
- [20] I. Fridovich, Superoxide radical and superoxide dismutases, Annu. Rev. Biochem. 64 (1995) 97–112.
- [21] R.N. Armstrong, Glutathione s-transferases: reaction mechanism, structure, and function, Chem. Res. Toxicol. 4 (1991) 10.
- [22] B. Leiers, A. Kampkotter, C.G. Grevelding, C.D. Link, T.E. Johnson, K. Henkle-Duhrsen, A stress-responsive glutathione s-transferase confers resistance to oxidative stress in *Caenorhabditis elegans*, Free Radic. Biol. Med. 34 (2009) 10.
- [23] E.M. Feder, G.E. Hofmann, Heat-shock proteins, molecular chaperones, and the stress response: evolutionary and ecological physiology, Annu. Rev. Physiol. 61 (1999) 243–282.
- [24] S. Franzellitti, E. Fabbri, Differential Hsp70 gene expression in the Mediterranean mussel exposed to various stressors, Biochem. Biophys. Res. Commun. 336 (2005) 1157–1163.
- [25] F.U. Hartl, A. Bracher, H.M. Hayer, Molecular chaperones in protein folding and proteostasis, Nature 475 (2011) 324–332.
- [26] F. Li, J. Xiang, Recent advances in researches on the innate immunity of shrimp in China, Dev. Comp. Immunol. 39 (2013) 11–26.
- [27] H. Guo, K. Li, W. Wang, C. Wang, Y. Shen, Effects of copper on hemocyte apoptosis, ros production, and gene expression in white shrimp *litopenaeus*

- vannamei*, Biol. Trace Elem. Res. 179 (2017) 318–326.
- [28] K.J. Livak, T.D. Schmittgen, Analysis of relative gene expression data using real-time quantitative PCR and the 2- $\Delta\Delta$ CT method, Methods 25 (2001) 402–408.
- [29] H.A.S. Sahul, M. Sarathi, R. Sudhakaran, G. Balasubramanian, M.S. Syed, Quantitative assessment of apoptotic hemocytes in white spot syndrome virus (wssv)-infected *penaeid shrimp*, *penaeus monodon* and *penaeus indicus*, by flow cytometric analysis, Aquaculture 256 (2006) 111–120.
- [30] D.R. Green, J.C. Reed, Mitochondria and apoptosis, Science 281 (1998) 1309.
- [31] D.L. McCulloch, Metabolic response of the grass shrimp *Palaemonetes kadiakensis*, Aquat. Toxicol. 17 (1990) 263–274.
- [32] W.N. Wang, J. Zhou, P. Wang, T.T. Tian, Y. Zheng, Y. Liu, et al., Oxidative stress, DNA damage and antioxidant enzyme gene expression in the Pacific white shrimp, *Litopenaeus vannamei* when exposed to acute pH stress, Comp. Biochem. Physiol. C 150 (2009) 428–435.
- [33] A. Bianchini, J.M. Monserrat, Effects of methyl parathion on *Chasmagnathus granulatus* hepatopancreas: protective role of sesamol, Ecotoxicol. Environ. Saf. 67 (2007) 100–108.
- [34] J. Zhou, W.N. Wang, A.L. Wang, W.Y. He, Q.T. Zhou, Y. Liu, et al., Glutathione S-transferase in the white shrimp *Litopenaeus vannamei*: characterization and regulation under pH stress. Comp. Biochem. Physiol. C 150 (2009) 224–230.
- [35] Y.Y. Chen, J.C. Chen, K.C. Tseng, Y.C. Lin, C.L. Huang, Activation of immunity, immune response, antioxidant ability, and resistance against *Vibrio alginolyticus* in white shrimp *Litopenaeus vannamei* decrease under long-term culture at low pH, Fish Shellfish Immun. 46 (2015) 192–199.
- [36] J.H. Ming, J. Xie, P. Xu, X.P. Ge, W.B. Liu, J.Y. Ye, Effects of emodin and vitamin C on growth performance, biochemical parameters and two Hsp70s mRNA expression of Wuchang bream (*Megalobrama amblycephala* Yih) under high temperature stress, Fish Shellfish Immun. 32 (2012) 651–661.
- [37] B. Liu, J. Xie, X. Ge, P. Xu, A. Wang, Y. He, et al., Effects of anthraquinone

extract from *Rheum officinale* Bail on the growth performance and physiological responses of *Macrobrachium rosenbergii* under high temperature stress, Fish Shellfish Immun. 29 (2010) 49–57.

[38] N. Sheikhzadeh, N.H. Tayefi, O.A. Khani, E.M.H. Najafi, Effects of *Haematococcus pluvialis* supplementation on antioxidant system and metabolism in rainbow trout (*Oncorhynchus mykiss*), Fish physiol. Biochem. 38 (2012) 413–419.

[39] T. Olsen, L. Ellerbeck, T. Fisher, A. Callaghan, M. Crane, Variability in acetylcholinesterase and glutathione S-transferase activities in *Chironomus riparius* Meigen deployed in situ at uncontaminated field sites, Environ. Toxicol. Chem. 20 (2001) 1725–1732.

[40] L.L. Amado, C.E. Rosa, A.M. Leite, L. Morase, W.V. Pires, P.G.L. Leaes, et al., Biomarkers in croakers *Micropogonias furnieri* (Teleostei: Sciaenidae) from polluted and nonpolluted areas from the Patos Lagoon estuary (Southern Brazil): evidences of genotoxic and immunological effects, Mar. Pollut. Bull. 52 (2006) 199–206.

[41] B. Blanchette, X. Feng, B.R. Singh, Marine glutathione S-transferases, Mar. Biotechnol. 9 (2007) 513–542.

[42] N. Mcloughlin, D. Yin, L. Maltby, R. Wood, M. Yu, Evaluation of sensitivity and specificity of two crustacean biochemical biomarkers, Environ. Toxicol. Chem. 19 (2000) 2085–2092.

[43] P. Hoarau, B.M. Gnassia, M. Romeo, J.P. Girard, Differential induction of glutathione s-transferases in the clam *Ruditapes decussatus* exposed to organic compounds. Environ. Toxicol. Chem. 20 (2001) 523–529.

[44] J. Zhou, W. Lei, X. Yu, W.N. Wang, W.Y. He, A.L. Wang, et al., Effect of temperature on antioxidant enzyme gene expression and stress protein response in white shrimp, *litopenaeus vannamei*, J. Therm. Biol. 35 (2010) 284–289.

[45] E.E. Deane, N.Y. Woo, Cloning and characterization of the hsp70 multigene family from silver sea bream: modulated gene expression between warm and cold temperature acclimation. Biochem. Bioph. Res. Co. 330 (2005) 776.

[46] W.D. Jiang, Y. Liu, K. Hu, J. Jiang, S.H. Li, L. Feng, et al., Copper exposure induces oxidative injury, disturbs the antioxidant system and changes the nrf2/are

- (cuznsod) signaling in the fish brain: protective effects of myo-inositol. *Aquat. Toxicol.* 155 (2014) 301.
- [47] Z. Qian, X. Liu, L. Wang, X. Wang, Y. Li, J. Xiang, et al., Gene expression profiles of four heat shock proteins in response to different acute stresses in shrimp, *Litopenaeus vannamei*. *Comp. Biochem. Physiol. C* 156 (2012) 211–220.
- [48] T.D.P. Parrilla, S.T. Zenteno, Antioxidant enzyme activities in Pacific white shrimp (*Litopenaeus vannamei*) in response to environmental Hypoxia and reoxygenation, *Aquaculture* 318 (2011) 379–383.
- [49] R.M. Phillips, T.H. Ward, Influence of extracellular pH on the cytotoxicity and DNA damage of a series of indolequinone compounds, *Anticancer Res.* 21 (2001) 1795–1801.
- [50] F. Geret, A. Jouan, V. Turpin, M.J. Bebianno, R.P. Cosson, Influence of metal exposure on metallothionein synthesis and lipid peroxidation in two bivalve mollusks: the oyster (*Crassostrea gigas*) and the mussel (*Mytilus edulis*), *Aquat. Living Resour.* 15 (2002) 61–66.
- [51] X.L. Liu, Q.Y. Xi, L. Yang, H.Y. Li, Q.Y. Jiang, G. Shu, et al., The effect of dietary panax ginseng polysaccharide extract on the immune responses in white shrimp, *Litopenaeus vannamei*, *Fish. Shellfish Immunol.* 30 (2011) 495–500.
- [52] Y.R. Saadat, N. Saeidi, S.Z. Vahed, A. Barzegari, J. Barar, An update to dna ladder assay for apoptosis detection. *Bioimpacts Bi* 5 (2015) 25.
- [53] Y. Wang, J. Li, P. Liu, J. Li, Z. Zhang, Z. Chang, et al., The responsive expression of a caspase gene in Chinese shrimp *Fenneropenaeus chinensis* against pH stress, *Aquac. Res.* 42 (2011) 1214–1230.
- [54] J. Qiang, H. Yang, H. Wang, P. Xu, Z.L. Qi, J. He, Studies on blood biochemical indices and expression of hepatic HSP70 mRNA of different tilapia strains artificially challenged with *Streptococcus iniae*, *J. Fish. China* 36 (2012) 958–968.
- [55] H.R. Köhler, C. Bartussek, H. Eckwert, K. Farian, S. Gränzer, T. Knigge, et al., The hepatic stress protein (hsp70) response to interacting abiotic parameters in fish exposed to various levels of pollution, *J. Aquat. Ecosyst. Stress Recov.* 8 (2001) 261–279.

[56] J. Wan, X. Ge, B. Liu, J. Xie, S. Cui, M. Zhou, et al., Effect of dietary vitamin c on non-specific immunity and mrna expression of three heat shock proteins (hsps) in juvenile *megalobrama amblycephala*, under ph stress, Aquaculture 434 (2014) 325–333.

[57] C.C. Li, J.C. Chen, The immune response of white shrimp *Litopenaeus vannamei* and its susceptibility to *Vibrio alginolyticus* under low and high pH stress, Fish Shellfish Immun. 25 (2008) 701–709.

[58] J. Wan, X. Ge, B. Liu, J. Xie, S. Cui, M. Zhou, et al., Effect of dietary vitamin c on non-specific immunity and mrna expression of three heat shock proteins (hsps) in juvenile *megalobrama amblycephala*, under pH stress, Aquaculture 434 (2014) 325–333.

Figure legends

Fig. 1. pH levels in the seawater of tanks during the experimental period.

Fig. 2. Survival rate of shrimp during the experimental period. Each bar represents the mean value from three repetitions with standard error (SE). * Indicates a significant ($P < 0.05$) difference compared with the NH group.

Fig. 3. ROS production (A) in the hepatopancreas of shrimp during the experimental period; ROS production (B) in the midgut of shrimp during the experimental period. See Fig. 2 for statistical information.

Fig. 4. Relative expression of *MnSOD* (A) in the hepatopancreas of shrimp during the experimental period; Relative expression of *MnSOD* (B) in the midgut of shrimp during the experimental period. See Fig. 2 for statistical information.

Fig. 5. Relative expression of *GPx* (A) in the hepatopancreas of shrimp during the experimental period; Relative expression of *GPx* (B) in the midgut of shrimp during the experimental period. See Fig. 2 for statistical information.

Fig. 6. Relative expression of *GST* (A) in the hepatopancreas of shrimp during the experimental period; Relative expression of *GST* (B) in the midgut of shrimp during the experimental period. See Fig. 2 for statistical

information.

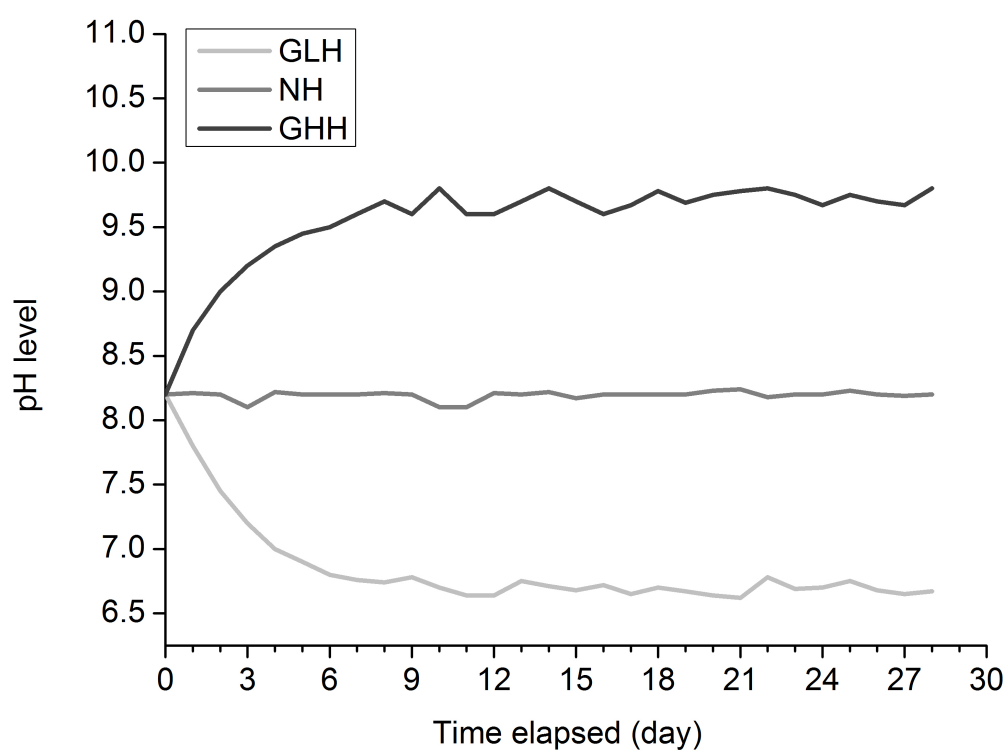
Fig. 7. Relative expression of *Hsp70* (A) in the hepatopancreas of shrimp during the experimental period; Relative expression of *Hsp70* (B) in the midgut of shrimp during the experimental period. See Fig. 2 for statistical information.

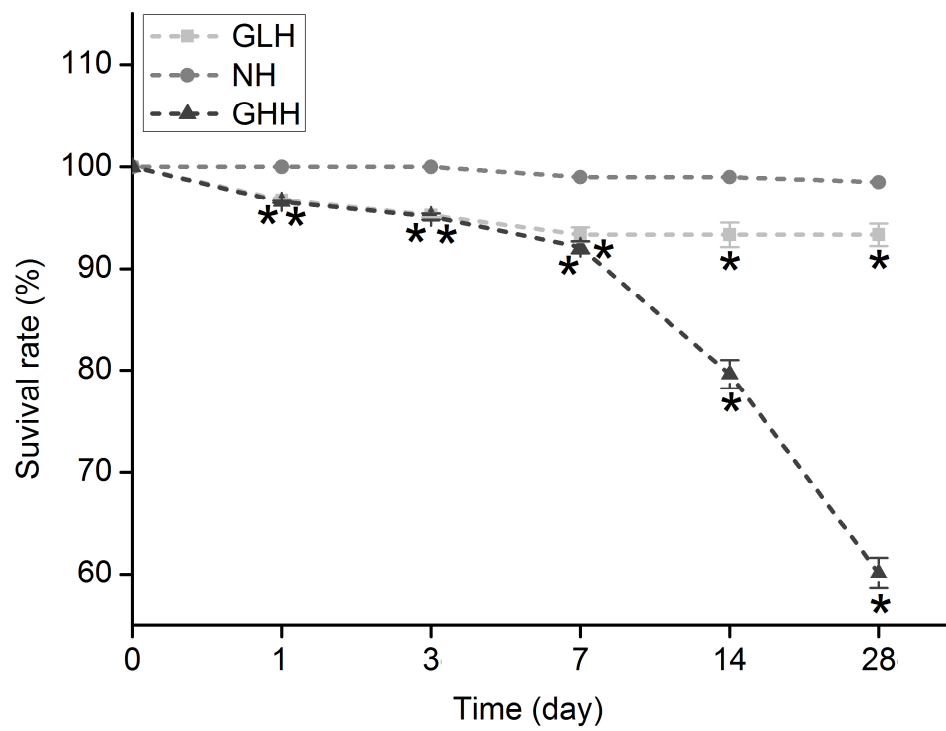
Fig. 8. MDA content (A) in the hepatopancreas of shrimp during the experimental period; MDA content (B) in the midgut of shrimp during the experimental period. See Fig. 2 for statistical information.

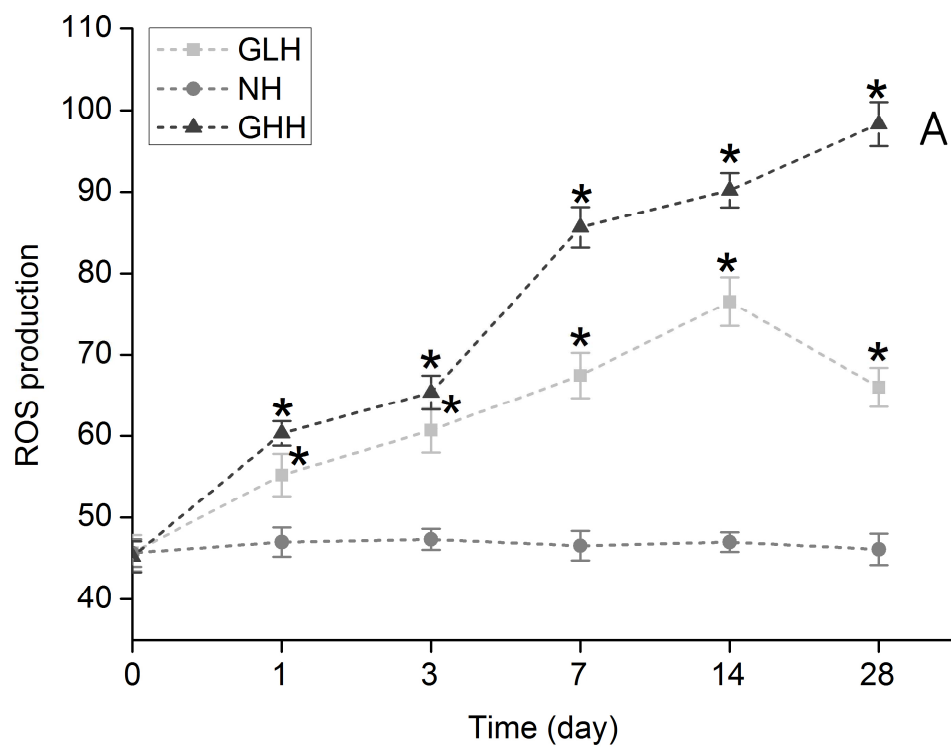
Fig. 9. DNA ladder in the hepatopancreases of GLH shrimp (A1), NH shrimp (B1), and GHH shrimp (C1); DNA ladder in the midgut of GLH shrimp (A2), NH shrimp (B2), and GHH shrimp (C2).

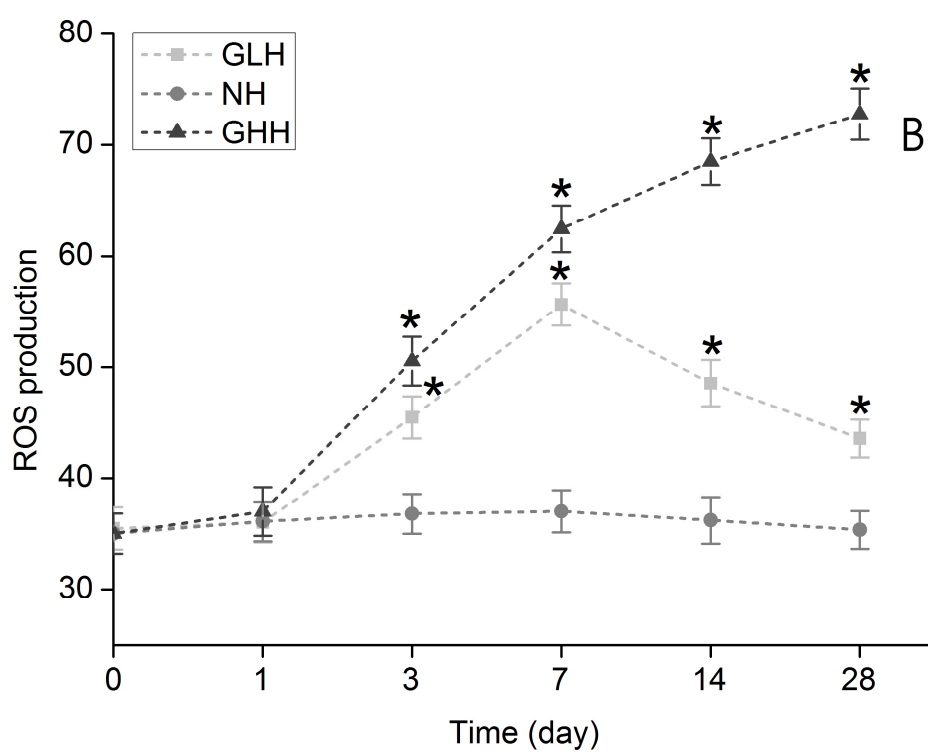
Table 1. Primers for the genes encoding *MnSOD*, *Gpx*, *GST*, *Hsp70* and β -actin in shrimp.

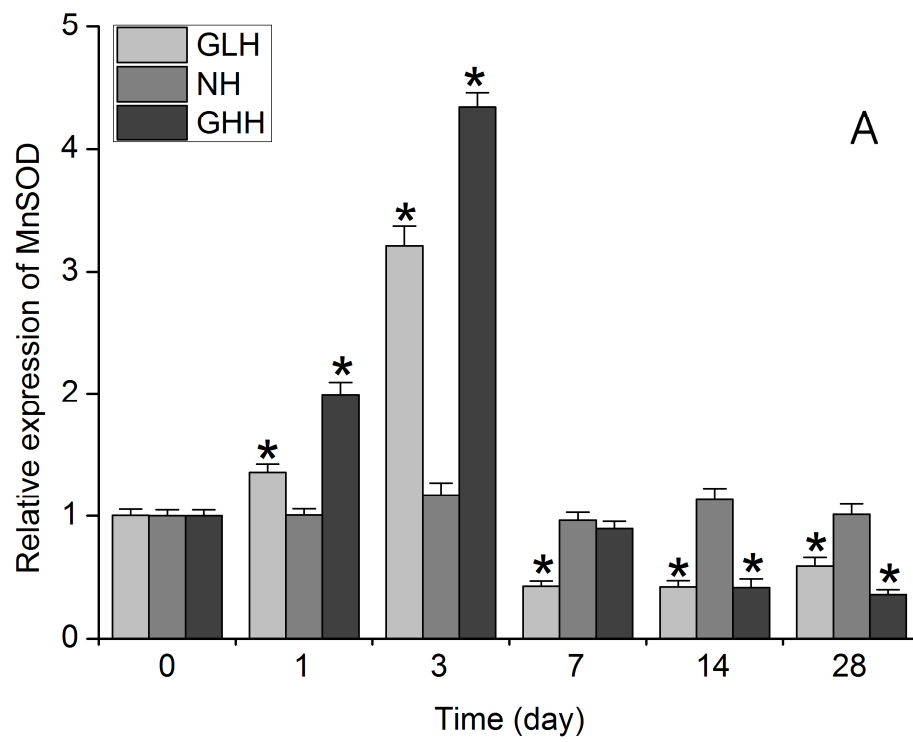
Gene	Nucleotide sequence (5'→3')	Amplicon (bp)
<i>MnSOD</i>	F-TGACGAGAGCTTTGGATCATTCC R-TGATTTGCAAGGGATCCTGGTT	156
<i>Gpx</i>	F-TTTATGTCAGATCCAAAGTGCATCA R-CAGCAAGTTTGCGATTTTCATCTTTA	149
<i>GST</i>	F- AAGATAACGCAGAGCAAGG R- TCGTAGGTGACGGTAAAGA	156
<i>Hsp70</i>	F-CTCCTGCGTGGGTGTGTT R-GCGGCGTCACCAATCAGA	186
β -actin	F- GCCCATCTACGAGGGATA R-GGTGGTCGTGAAGGTGTAA	121

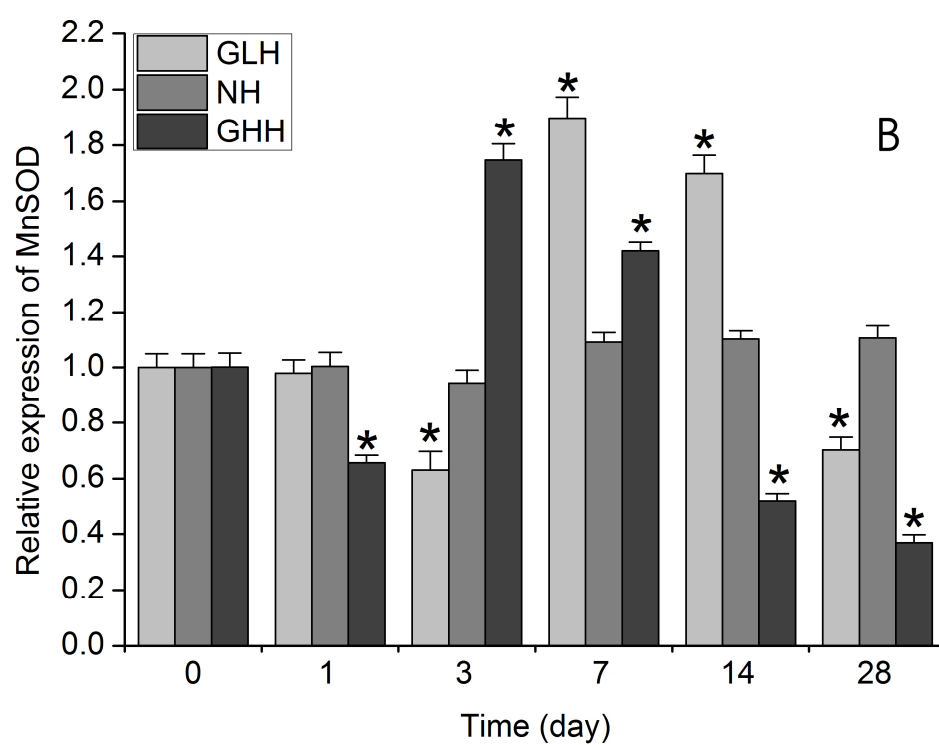


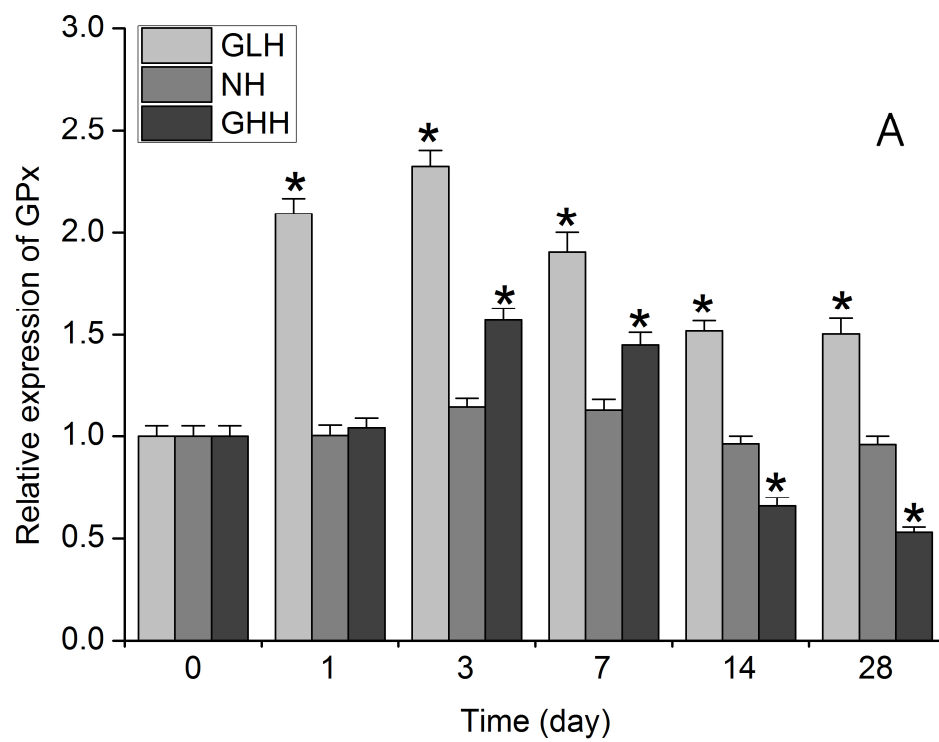


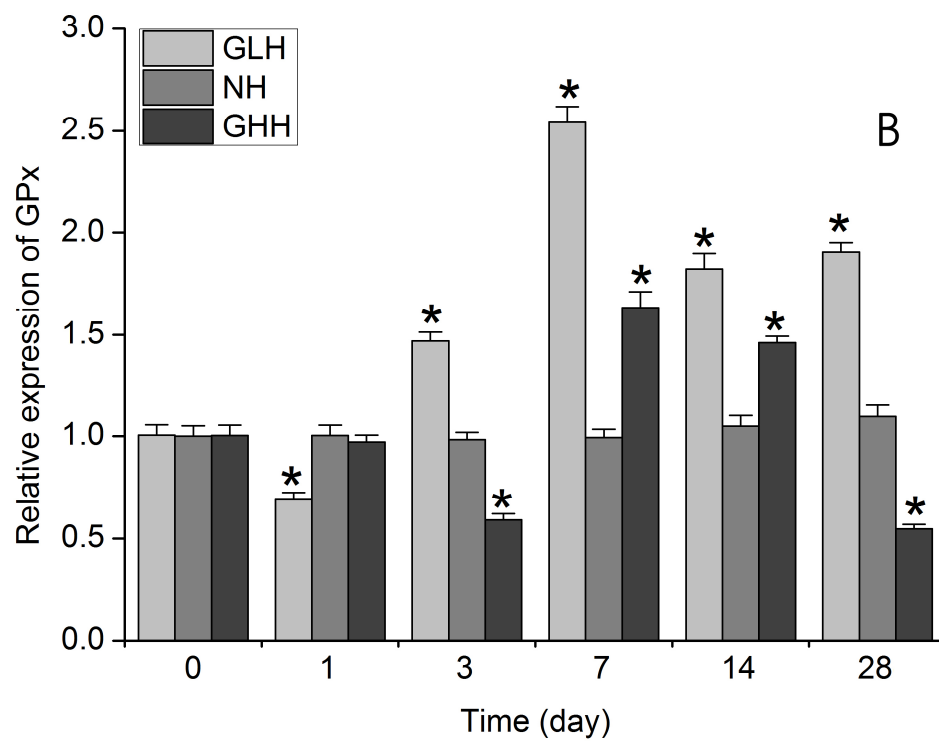


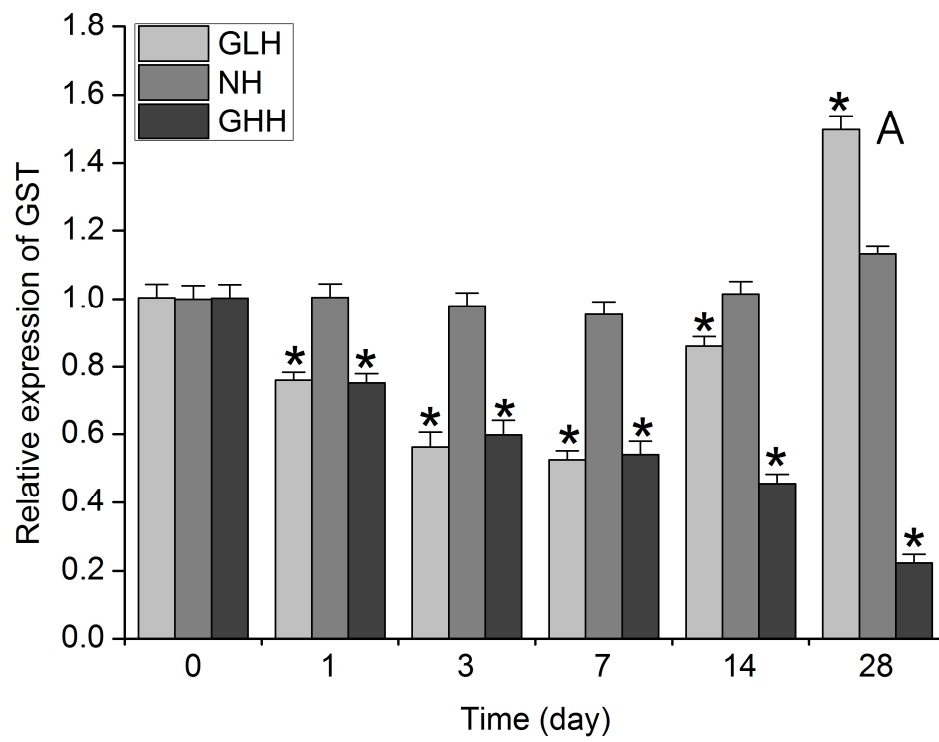


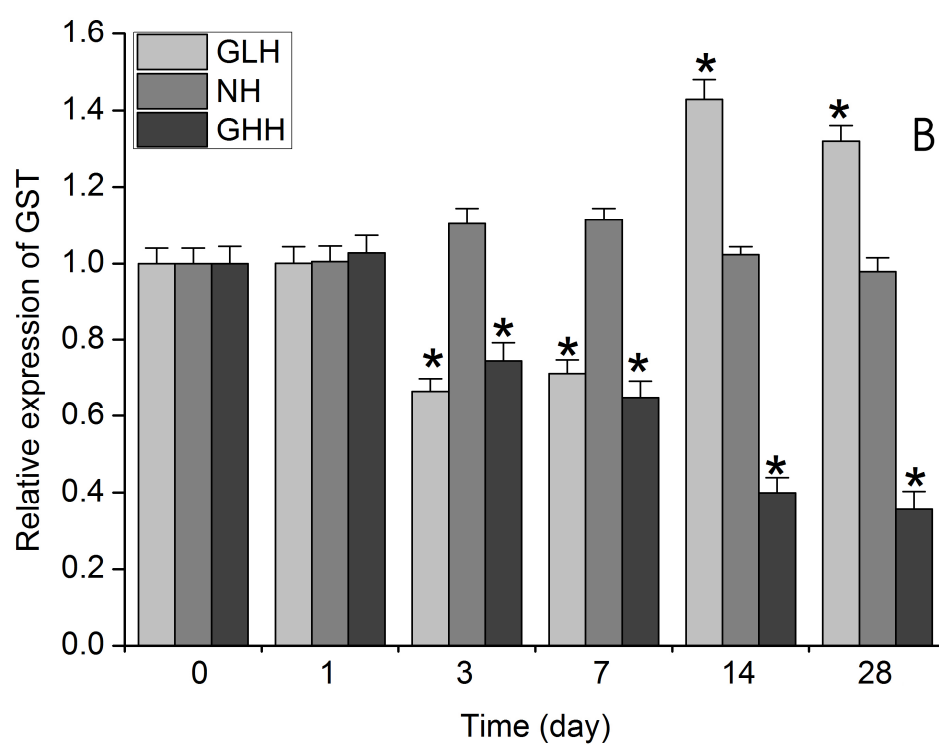


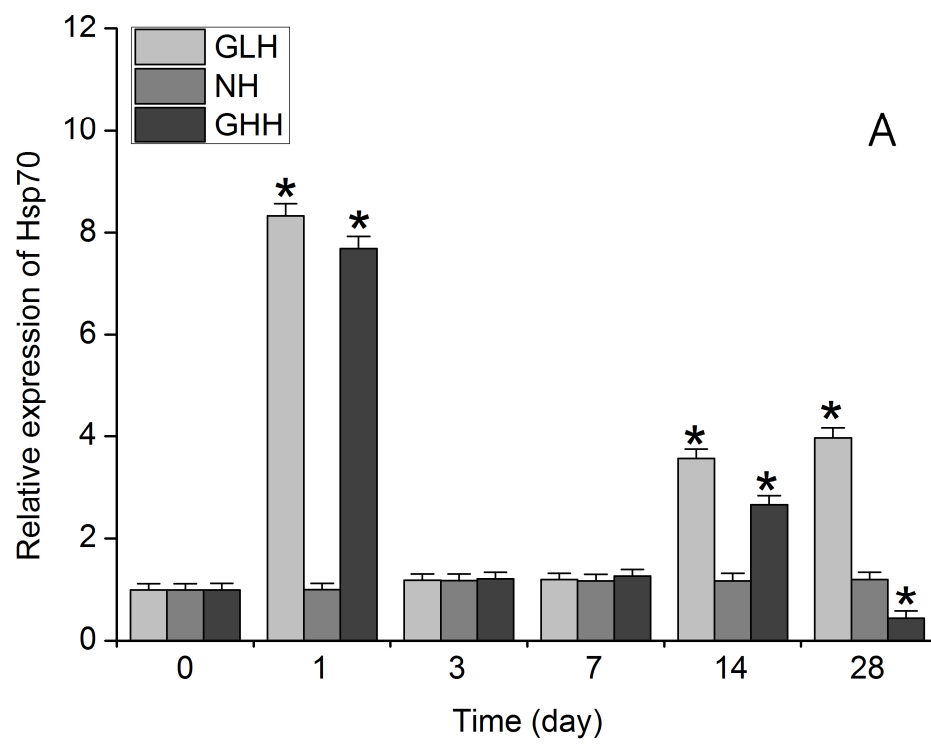


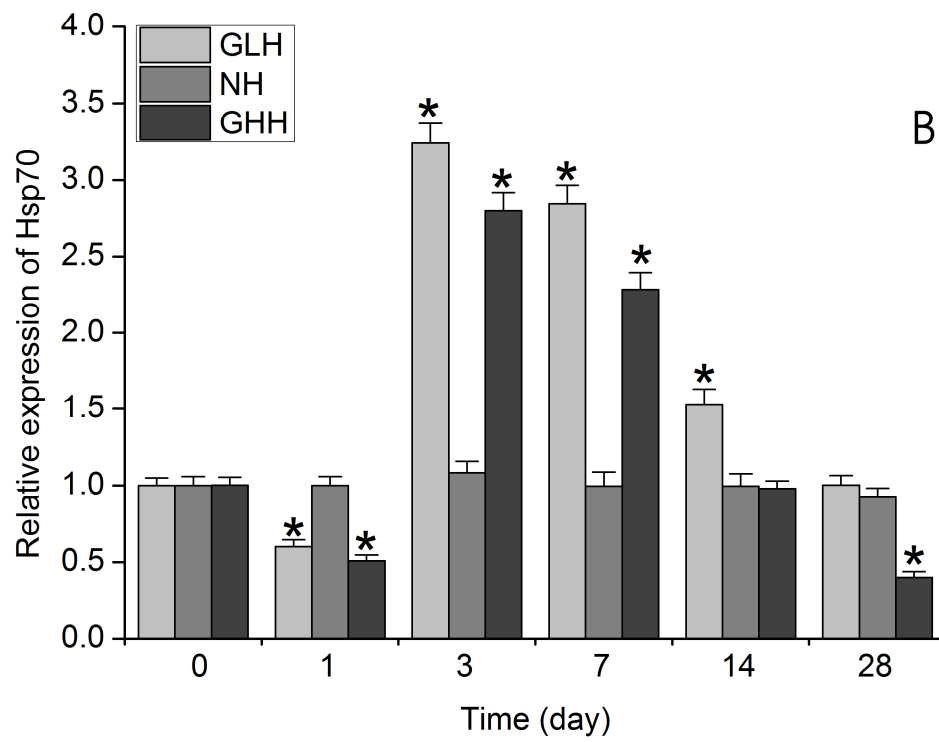


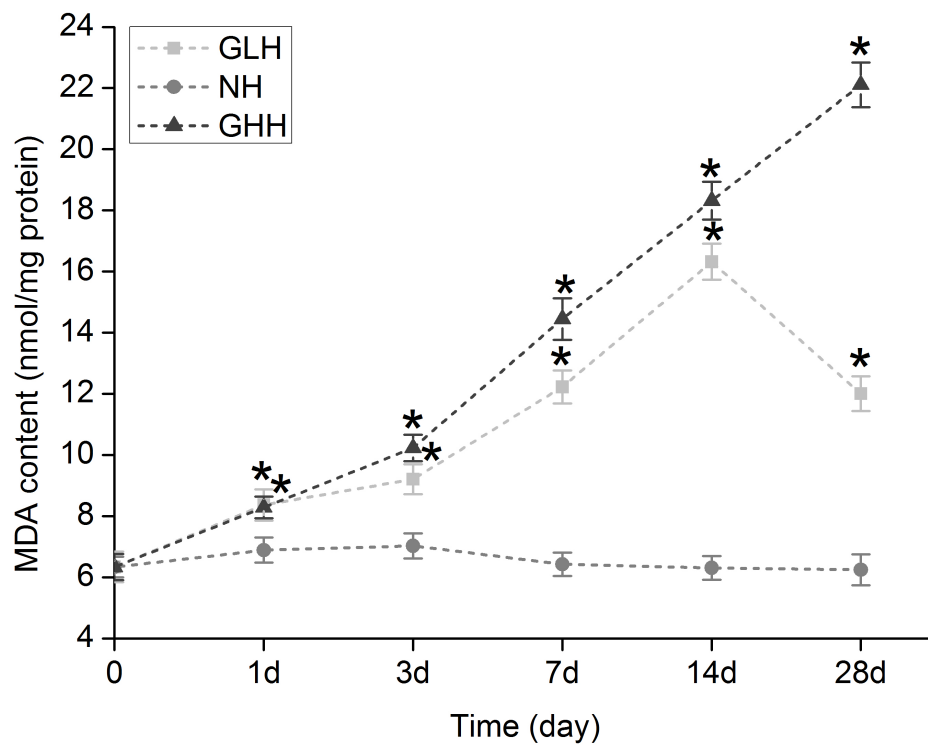


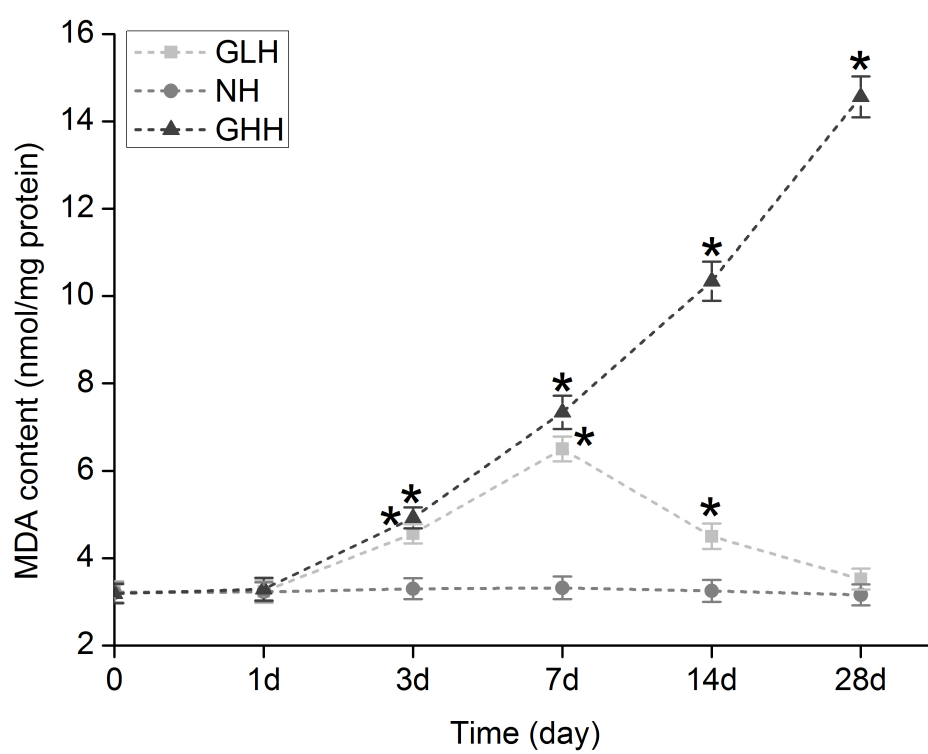


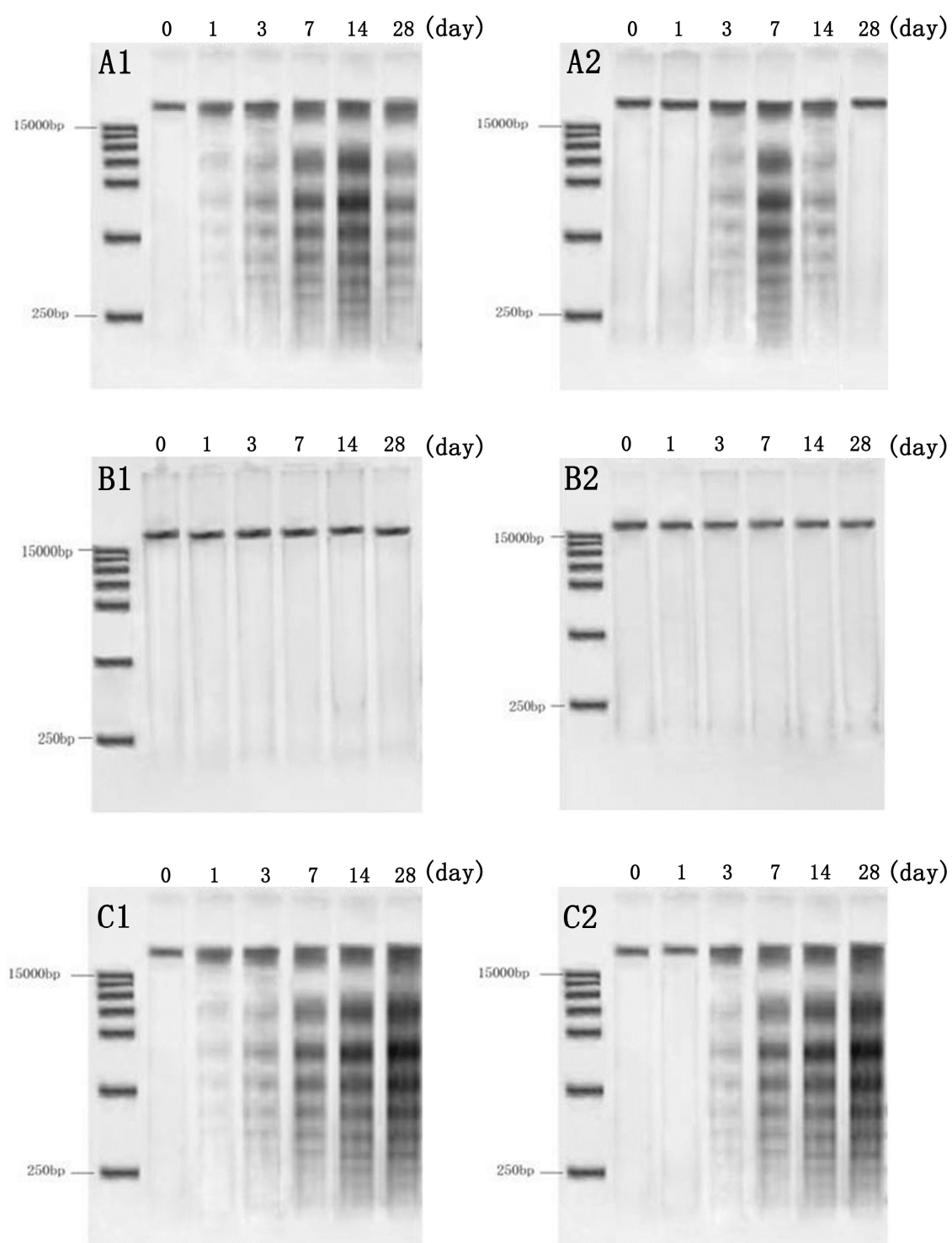












Highlights

1. The hepatopancreas in the *Litopenaeus vannamei* was more sensitive to ROS than midgut under gradual-low and high pH stress.
2. Shrimp could lose antioxidant regulatory capacity in the hepatopancreas and the midgut after long-term gradual-high pH stress.
3. Enhanced antioxidant response in the hepatopancreas and the midgut might be long-term antioxidant adaptation mechanism of shrimp to gradual-low pH stress.
4. These results help us to realize that it is necessary to protect hepatopancreas for controlling shrimp death under gradual-high pH stress.